# S13 — Do people register the centroid of ignored distractors?

**Question.** While tracking four target dots, can people point to the centroid of the four *distractor* dots when those distractors vanish, more accurately than the empirical chance level from a guessing phase? (Directional: test-phase error < guessing error.)

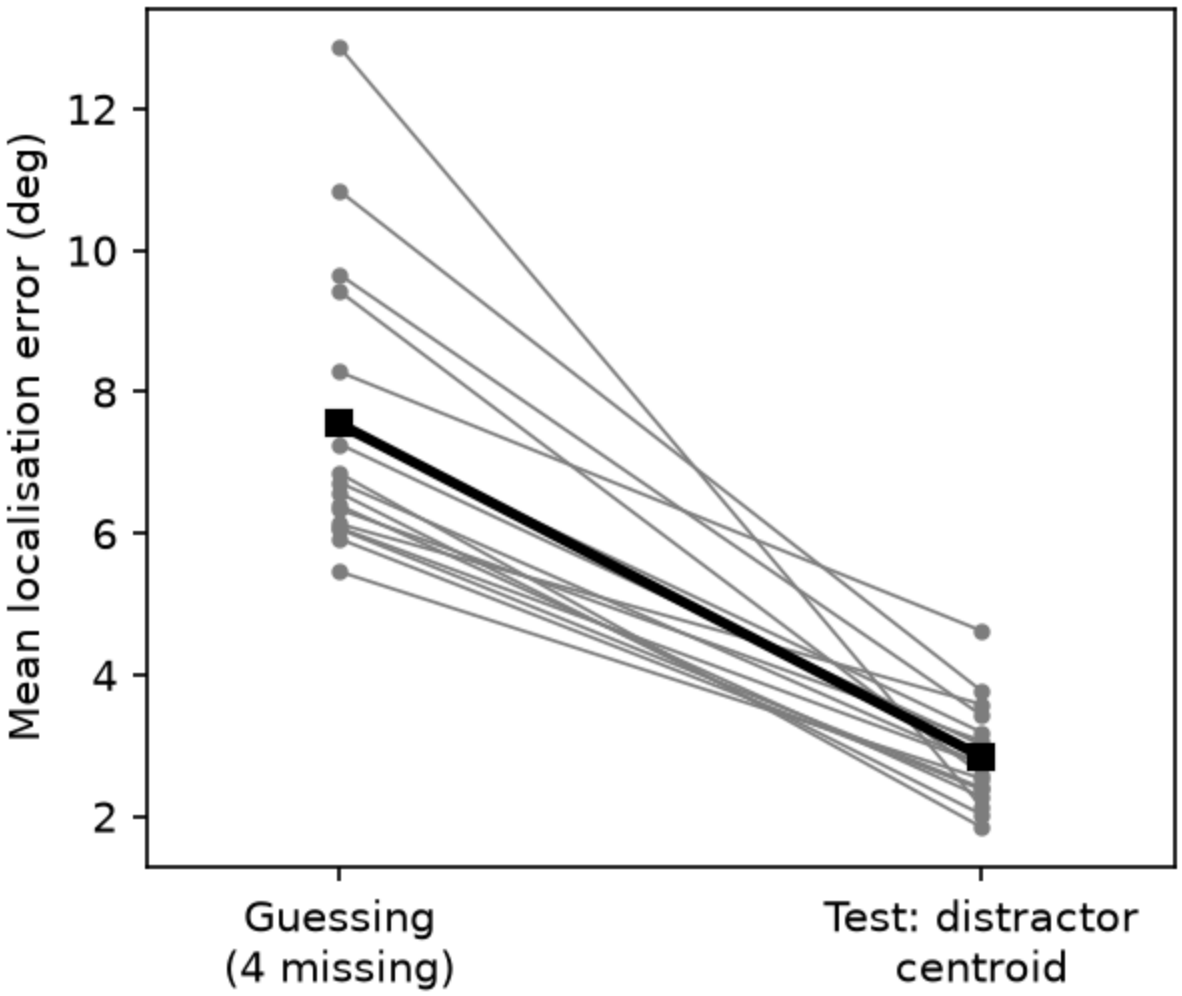
**Data and sample.** Replication file `d19.dta`: 17 participants × 120 trials (80 test, 40 guessing), 2,040 trials. Outcome: `angle`, localisation error in degrees of visual angle (computed in the data file from click and true location; for four missing dots, the area-weighted centroid). No participants excluded (as specified), no trials excluded (no missing values), no trimming. **n = 17.** Each participant contributes 20 distractor-centroid test trials and 20 four-missing guessing trials.

**Coding note.** The file does not label which `tar_vs_dis` code means "distractor". Before looking at any centroid error, I fixed a rule: the code with the larger *single-dot* test-phase error is the distractor (tracked targets must be localised better). Code 11 had larger single-dot error than code 1 (8.52° vs 6.32°, t(16) = −5.20, p < .001), so code 11 = distractor. The primary result is also reported under the reversed mapping: it does not change the conclusion.

**Primary test (pre-specified).** Per participant, mean error on distractor-centroid test trials minus mean error on four-missing guessing trials; paired t-test, two-tailed; dz = t/√n, 95% CI by noncentral t.

| | **Mean error (deg)** |
|---|---|
| Test phase, distractor centroid | 2.85 |
| Guessing phase, four missing (chance) | 7.55 |
| Difference (test – guess) | **−4.70, 95% CI [−5.74, −3.66]** |

**$t(16) = -9.59$, $p = 4.9 \times 10^{-8}$, $dz = -2.33$, 95% CI [−3.25, −1.39].** All 17 participants were more accurate in the test phase than when guessing (Wilcoxon W = 0, $p = 1.5 \times 10^{-5}$).




**Robustness and secondary results (not the basis of the conclusion).** - Per-participant medians instead of means: $t(16) = -8.85$, $p = 1.5 \times 10^{-7}$, $dz = -2.15$. - Reversed code mapping (code 1 as distractor): diff = −3.80°, $t(16) = -7.49$, $p = 1.3 \times 10^{-6}$, $dz = -1.82$. - Single distractor vs single-dot guessing: 8.52° vs 9.95°, $t(16) = -3.65$, $p = .002$, $dz = -0.89$. - Target centroid vs guessing: 3.75° vs 7.55°, $t(16) = -7.49$, $p < .001$. Target centroid error was *larger* than distractor centroid error (3.75° vs 2.85°, $t(16) = 4.54$, $p < .001$). - Guessing displays generated as target-type vs distractor-type did not differ reliably (7.94° vs 7.16°, $p = .14$).

**Caveats.** (1) The target/distractor coding is inferred from the data (by a rule fixed in advance), not given; the conclusion holds under either coding. (2) The "chance" baseline is the guessing phase, which came as a separate block without the tracking task; differences in motivation or strategy between phases could add to the gap, though a gap of 4.7° with every participant in the same direction is unlikely to be explained by that alone. (3) The finding that distractor centroids were localised better than target centroids is unexpected; I report it without explanation.

**Conclusion.** The claim holds on this data: localisation error for the centroid of the missing distractors (2.85°) was far smaller than the empirical chance error (7.55°), $t(16) = -9.59$, $p < .001$, $dz = -2.33$.